



Entergy Operations, Inc.
P. O. Box 756
Port Gibson, MS 39150

Eric A. Larson
Site Vice President
Grand Gulf Nuclear Station
Tel. (601) 437-7500

10CFR50.73

GNRO-2018/00018

April 17, 2018

U.S. Nuclear Regulatory Commission
Attn: Document Control Desk
Washington, DC 20555-0001

SUBJECT: Licensee Event Report 2018-004-00, Potential Loss of Safety Function Due to Concurrent Inoperability of Two Diesel Generators
Grand Gulf Nuclear Station, Unit 1
Docket No. 50-416
License No. NPF-29

Dear Sir or Madam:

Attached is Licensee Event Report 2018-004-00, Potential Loss of Safety Function Due to Concurrent Inoperability of Two Diesel Generators. This report is being submitted in accordance with 10CFR50.73(a)(2)(v)(D) as an event or condition that could have prevented the fulfillment of the safety function.

This letter contains no new commitments. If you have any questions or require additional information, please contact Douglas Neve at 601-437-2103.

Sincerely,

A handwritten signature in black ink, appearing to read "E. A. Larson", written over a horizontal line.

Eric A. Larson
Site Vice President
Grand Gulf Nuclear Station
EAL/tdf

Attachment: Licensee Event Report 2018-004-00

cc: see next page

U.S. Nuclear Regulatory Commission
ATTN: Mr. Siva Lingam
Mail Stop OWFN 8 B1
Rockville, MD 20852-2738

NRC Senior Resident Inspector
Grand Gulf Nuclear Station
Port Gibson, MS 39150

U. S. Nuclear Regulatory Commission
ATTN: Mr. Kriss Kennedy, NRR/DORL (w/2)
Mail Stop OWFN 8 B1
Washington, DC 20555-0001

Attachment

Licensee Event Report (LER) 2018-0040-00



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. Facility Name Grand Gulf Nuclear Station	2. Docket Number 05000416	3. Page 1 OF 4
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4. Title Potential Loss of Safety Function Due to Concurrent Inoperability of Two Diesel Generators
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5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
02	18	2018	2018	- 004	- 00	04	17	2018	N/A	05000 N/A
									Facility Name	Docket Number
									N/A	05000 N/A

9. Operating Mode Mode 1	11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)							
	☐ 20.2201(b)		☐ 20.2203(a)(3)(i)		☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)	
	☐ 20.2201(d)		☐ 20.2203(a)(3)(ii)		☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)	
	☐ 20.2203(a)(1)		☐ 20.2203(a)(4)		☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)	
10. Power Level 97	☐ 20.2203(a)(2)(i)		☐ 50.36(c)(1)(i)(A)		☐ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)	
	☐ 20.2203(a)(2)(ii)		☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(v)(A)		☐ 73.71(a)(4)	
	☐ 20.2203(a)(2)(iii)		☐ 50.36(c)(2)		☐ 50.73(a)(2)(v)(B)		☐ 73.71(a)(5)	
	☐ 20.2203(a)(2)(iv)		☐ 50.46(a)(3)(ii)		☐ 50.73(a)(2)(v)(C)		☐ 73.77(a)(1)	
	☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(A)		☒ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(ii)	
	☐ 20.2203(a)(2)(vi)		☐ 50.73(a)(2)(i)(B)		☐ 50.73(a)(2)(vii)		☐ 73.77(a)(2)(iii)	
		☐ 50.73(a)(2)(i)(C)		☐ Other (Specify in Abstract below or in NRC Form 366A				

12. Licensee Contact for this LER

Licensee Contact

Douglas A. Neve

Telephone Number (Include Area Code)

601.437.2103

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable To ICES	Cause	System	Component	Manufacturer	Reportable To ICES
B	BG	TS	Barksdale	Yes	N/A	N/A	N/A	N/A	N/A

14. Supplemental Report Expected

☐ Yes (If yes, complete 15. Expected Submission Date) ☒ No

15. Expected Submission Date

Month	Day	Year
N/A	N/A	N/A

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

On February 14, 2018, the Grand Gulf Nuclear Station Division 2 diesel generator (DG) was removed from service for maintenance. On February 18, 2018, at 0006 hours, while in Mode 1 at 97 percent power, the Division 3 DG experienced a lockout and was declared inoperable. Technical Specification 3.8.1 Condition E was entered for two DGs inoperable at that time. On February 18, 2018, at 0355 hours, Division 2 DG maintenance was completed, the diesel was returned to service, and Condition E was exited. Division 3 DG was restored to operable status on February 18, 2018, at 1257 hours.

The Division 3 DG lockout was caused by inadequate operation procedures, which failed to detect an elevated jacket water temperature condition. The elevated jacket water temperature was caused by a malfunctioning jacket water temperature control switch. Corrective actions include replacement of the temperature control switch, and revision of the operation procedures.

Division 1 DG, which powers one low pressure core injection pump and one low pressure core spray pump, remained operable. An evaluation of Chapter 15 accident analyses demonstrated that, with just Division 1, peak cladding temperatures and cladding oxidation remained within the Cycle 21 licensing basis of <1730 degrees F and <3%, respectively.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Grand Gulf Nuclear Station	05000-00416	2018	- 004	- 00

NARRATIVE**A. PLANT CONDITIONS PRIOR TO THE EVENT**

Prior to the event Grand Gulf Nuclear Station (GGNS) Unit 1 was operating in Mode 1 at approximately 97 percent power in steady state conditions. The division 2 diesel generator (DG) [EIS: DG] was inoperable due to preplanned maintenance and contributed to the event. The division 2 electrical service was being supplied with offsite power and remained available and functional. There were no other structures, systems, or components that were known to be inoperable that contributed to the event.

B. DESCRIPTION

On February 17, 2018, at 2353 hours, the high pressure core spray (HPCS) diesel [EIS: BG][DG] trouble alarm was received in the control room. At 0006 hours, while operators were in route to the diesel, control room operators received diesel engine trip and not ready for start alarms. Investigation revealed that the Division 3 HPCS DG was tripped and not available for autostart. Division 3 DG was declared inoperable and Technical Specification 3.8.1 Condition E, for two required DGs inoperable, was entered on February 18, 2018, at 0006 hours. Division 3 electrical service was being supplied with offsite power and remained available and functional.

Operations personnel detected a local jacket water temperature alarm due to high jacket water temperature. Operators assumed manual control of the Division 3 diesel jacket water temperature, restored the temperature to an acceptable level, cleared the Division 3 DG trips, and restored the DG to available status on February 18, 2018, at 0037 hours.

The Division 2 DG was returned to service and made operable on February 18, 2018 at 0355 hours. Technical Specification 3.8.1 Condition E was exited at that time.

The Division 3 diesel generator high jacket water temperature was caused by a malfunctioning temperature control switch [EIS: TS]. Investigation indicated the temperature switch had failed approximately 7 days prior to the event, and it took that long for the system temperature to reach trip setpoint. Operations procedures were inadequate to detect the temperature increase prior to reaching the trip setpoint. The switch was replaced and retested satisfactorily, and the Division 3 diesel generator was returned to operable status on February 18, 2018 at 1240 hours.

C. REPORTABILITY

This report is made pursuant to 10CFR50.73(a)(2)(v)(D) for any event or condition that could have prevented fulfillment of the safety function of structures or systems that are needed to mitigate the consequences of an accident.

This event was reported under 10CFR50.72(b)(3)(v)(D) in ENS notification 53227.

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		2018	- 004	- 00

D. CAUSE

- Operator rounds procedure contained non-conservative lube oil temperature limits and did not require monitoring of jacket water temperature.
- Neither the HPCS DG system's alarm response instruction nor the HPCS DG system operating instruction contained guidance on manually controlling jacket water heaters to maintain acceptable standby temperatures.
- The engineering System Monitoring Plan contained non-conservative lube oil temperature limits and did not require monitoring of jacket water temperature.
- Failure of jacket water temperature switch

E. CORRECTIVE ACTIONS

The following corrective actions are completed or planned.

Completed:

- The Division 3 DG jacket water temperature control switch was replaced and retested satisfactorily
- The lube oil temperature limits in the System Monitoring Plan were revised to be more conservative

Planned actions included in the corrective action program (which may be changed in accordance with the program):

- Division 3 DG jacket water temperature monitoring will be added to the operator rounds procedure
- DG lube oil and jacket water temperature limits in the operator rounds procedure will be revised to a more conservative range
- Guidance will be added to Division 3 DG operating procedures for manually controlling jacket water heaters to maintain acceptable standby temperatures
- Division 3 jacket water temperature will be added to the System Monitoring Plan
- System engineering will review the operator rounds procedures to ensure the proper safety system parameters are monitored and that the established limits are conservative with respect to identifying abnormalities prior to an unplanned loss of the system

F. SAFETY SIGNIFICANCE

Division 2 and 3 DGs were concurrently inoperable for approximately 4 hours. Division I DG, which powers one low pressure core injection (LPCI) pump and one low pressure core spray (LPCS) pump, remained operable during this event. An evaluation of large break and small break loss of coolant events demonstrated that, with just Division I, peak cladding temperature increases from 1632 degrees F to 1719 degrees F but remains below the Cycle 21 licensing basis of <1730 degrees F and well below the 2200 degree F limit in 10CFR50.46. Cladding oxidation remains below the 3% limit. The Division 3 DG was made functional after the initial 31 minutes and was available to further mitigate an event.

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The event posed no threat to the health and safety of the general public or to nuclear safety. No technical specification safety limits were violated. Industrial safety was not challenged, and there was no potential or actual radiological release during the event.

Therefore, the safety significance of this event is considered to be very low. This condition will not be counted as a safety system functional failure against Performance Indicator MS05, Safety System Functional Failures.

G. PREVIOUS SIMILAR EVENTS

Entergy conducted a three-year review of condition reports related to temperature monitoring of diesel generators. Two events were found upon review.

A 2015 event was an immersion heater that was stuck in the ON position but did not result in a trip, so it was not considered a failure and no cause evaluation was performed, but the switch was replaced and retested satisfactorily. The action taken in that event would not have prevented this event from occurring.

A 2017 event was the failure of division 3 diesel immersion heaters not being energized. This was caused by foreign material in the associated breaker preventing contacts to make up. The action taken in that event would not have prevented this event from occurring.